2019 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.361 Artificial Intelligence

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Simple calculators permitted

Question 1

1.a Describe the 5 levels of autonomy including the sub-levels for levels 4 and 5. Give an example of how each level should be interpreted in terms of controlling a car.

[9 marks]

1.b Describe the procedure of the k-means clustering algorithm.

[8 marks]

1.c For the following two data samples:

$x_1=\{9; -7; 5; 13\}$

$x_2=\{16; -42; 58; 4\}$

i. calculate the range and mean;

[4 marks]

ii. perform normalisation;

[4 marks]

Total 25 Marks

Question 2

2.a Describe the three main types of machine learning methods and give an example for each one of them.

[6 marks]

2.b Sketch and compare biological and computational neurons.

[5 marks]

2.c Describe and compare the two main types of fuzzy rule-based systems. Provide an example for each one of them.

[6 marks]

2.d In the following Table the expected profit of three investment decisions made by an investment banker under different conditions (future oil price changes) is given. Using Maximax, Maximin, Lagrangian (equal probabilities), and Minimum Regret Decision Making methods determine which investment is best and what it will bring to the investment banker.

Future oil price changes	**Expected Profit**		
	Transport & Travel	Manufacturing	Agriculture
Decrease	*£900M*	*£150M*	*£500M*
Stable	*£250M*	*£300M*	*£500M*
Increase	*£80M*	*£500M*	*£400M*

[8 marks]

Total 25 Marks

Question 3

3.a Describe three common types of chromosome encoding methods and give an example for each of them.

[6 marks]

3.b Given below is an example genetic algorithm with an initial population consisting of six chromosomes.

C_1	1010101010	C_2	1101101101
C_3	0001000100	C_4	0010010010
C_5	0000000000	C_6	1000000000

i. The fitness value is calculated as the number of 1s present in the chromosome. Calculate the fitness of each chromosome in the initial population.

[3 marks]

ii. If the Roulette Wheel Selection algorithm is used as the selection method, is there any chromosome that will never be selected? Which chromosome and why?

[4 marks]

iii. In order to select two parents, the Roulette Wheel Selection algorithm is used. Assume that two randomly generated numbers are 0.6 and 0.9. Fill in the following table and specify which chromosomes are selected as parents.

Chromosome	C_1	C_2	C_3	C_4	C_5	C_6
Fitness						
Probability						
Cumulative Probability						

[8 marks]

iv. In order to generate two new offsprings, the Uniform Crossover is used for recombination of the selected parent chromosomes in (iii). Assume that Uniform Crossover uses the following randomly generated numbers and the swapping probability is 0.5. What are the generated offsprings?

0.3	0.2	0.8	0.1	0.1	0.9	0.4	0.2	0.7	0.2

[4 marks]

Total 25 Marks

Question 4

4.a Describe the Naïve Bayes classifier. How does the Naïve Bayes classifier estimate probabilities from continuous attributes?

[4 marks]

4.b Take the dataset below that informs about decision making attributes to play tennis outside for different days. The dataset consists of 14 examples with four attributes ("Outlook", "Temperature", "Humidity", and "Wind") and a Binary outcome ("Play Tennis"). If you build an optimal decision tree according to the Information Gain (IG) score, which attribute would be chosen as the root node of the decision tree? In order to answer this question, calculate the Information Gain (IG) score for "Temperature" and "Humidity" attributes only and fill in the following table. The IG score of attribute A is: IG(A) = Entropy before – AvgEntropy after choosing A. You need to show your calculations, and you only need to do this for the root node of the decision tree.

Attribute	"Outlook"	"Temperature"	"Humidity"	"Wind"
Information Gain	0.246	?	?	0.0478

Outlook	Temperature	Humidity	Wind	Play Tennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

[15 marks]

Question 4 continues on next page...

Question 4 continued.

4.c Which attribute would be chosen as the root node of the decision tree? Why?

[2 marks]

4.d Consider the following confusion matrix of a classification system with two possible classes on 500 samples. Calculate accuracy, precision, recall, and F1-Score for this classification system.

		Predicted Class	
Actual Class		Class="Yes"	Class="No"
	Class="Yes"	210	40
	Class="No"	60	190

[4 marks]

Total 25 Marks

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